插值 $f(x_i)=\varphi(x_i)=\sum_{j=0}^{n}C_j\psi_j(x_i)$ $x_0,x_1,\cdots,x_n$ （插值条件 ↑ 插值函数式）

（↓ 被插值函数 ↓ 插值节点）

Lagrange插值 基函数 $l_k(x)=\prod_{i=0,i\neq k}^{n}\frac{x-x_i}{x_k-x_i}$ $l_k(x_i)=\delta_{ik}$ $L_n(x)=\sum_{k=0}^{n}f_k l_k(x)$

余项 $R_n(x)=f(x)-L_n(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)$（节点多项式 $\omega_{n+1}(x)$）

Newton插值 $N_{n-1}(x_i)=f_i,\ 0\le i\le n-1$ $N_n(x)=N_{n-1}(x)+q_n(x)$

$q_n(x_i)=N_n(x_i)-N_{n-1}(x_i)=0,\ 0\le i\le n-1 \Rightarrow q_n(x)=a_n(x-x_0)\cdots(x-x_{n-1})$

$a_n=\frac{f(x_n)-N_{n-1}(x_n)}{\omega_n(x_n)}=f[x_0,x_1,\cdots,x_n]$

$N_n(x)=\sum_{k=0}^{n}f[x_0,\cdots,x_k]\omega_k(x)$ $\omega_0=1$

$=f[x_0]+f[x_0,x_1](x-x_0)+f[x_0,x_1,x_2](x-x_0)(x-x_1)+\cdots$

$f[x_0,x_1,\cdots,x_n]=\frac{f[x_1,\cdots,x_n]-f[x_0,\cdots,x_{n-1}]}{x_n-x_0}$ $f[\overbrace{x_0,x_0,\cdots,x_0}^{m+1\text{阶}}]=\frac{f^{(m)}(x_0)}{m!}$

$R_n(x)=f(x)-N_n(x)=f[x,x_0,x_1,\cdots,x_n]\prod_{i=0}^{n}(x-x_i)=\frac{f^{(n+1)}(\xi(x))}{(n+1)!}\omega_{n+1}(x)$

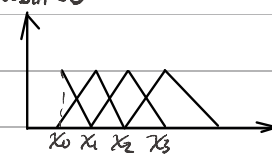
Hermite插值 $\alpha_i(x_j)=\delta_{ij}$, $\alpha_i'(x_j)=0$, $\beta_j(x_j)=0$ $\beta_i'(x_j)=\delta_{ij}$

$H_{2n+1}(x)=\sum_{i=0}^{n}[f_i\alpha_i(x)+f_i'\beta_i(x)] \Rightarrow H_{2n+1}(x_k)=f_k\ H_{2n+1}'(x_k)=f_k'$

$\alpha_i(x)=(ax+b)l_i^2(x)$ $\begin{cases}ax_i+b=1\\ a+2l_i'(x_i)=0\end{cases}$ $\beta_i(x)=(x-x_i)l_i^2(x)$

$R_{2n+1}(x)=f(x)-H_{2n+1}(x)=\frac{f^{(2n+2)}(\xi)}{(2n+2)!}\omega_{2n+1}^2(x)$

分段线性插值 $l_i(x)=\begin{cases}\frac{x-x_{i-1}}{x_i-x_{i-1}}, & x\in[x_{i-1},x_i]\\ \frac{x-x_{i+1}}{x_i-x_{i+1}}, & x\in[x_i,x_{i+1}]\end{cases}$



三次样条插值 $S_3(x_i)=f(x_i)$, $S_3^i(x_{i+1})=S_3^{i+1}(x_{i+1})$ $\frac{dS^i}{dx}(x_{i+1})=\frac{dS^{i+1}}{dx}(x_{i+1})\cdots$

固支条件：$S_3'(x_0)=f'(x_0)$, $S_3'(x_n)=f'(x_n)$

自然条件：$S_3''(x_0)=0$, $S_3''(x_n)=0$

周期条件：$S_3(x_0)=S_3(x_n)$、$S_3'(x_0)=S_3'(x_n)$ $S_3''(x_0)=S_3''(x_n)$

逼近 $\|p-f\| = \min_{S\in\Phi_n} \|S-f\|$ $\varphi_0, \varphi_1, \cdots, \varphi_n$ 线性无关 $S(x) = \sum_{j=0}^{n} a_j \varphi_j(x)$

最佳平方逼近: $\int_a^b \rho(x)[f(x) - \sum_{j=0}^{n} a_j \varphi_j(x)]^2 dx = \|f-S\|_2^2$

法方程 $\sum_{j=0}^{n} (\varphi_j, \varphi_k) a_j = (f, \varphi_k)$

$$\begin{bmatrix} (\varphi_0,\varphi_0) & (\varphi_0,\varphi_1) & \cdots & (\varphi_0,\varphi_n) \\ (\varphi_1,\varphi_0) & (\varphi_1,\varphi_1) & \cdots & (\varphi_1,\varphi_n) \\ \vdots & & \ddots & \vdots \\ (\varphi_n,\varphi_0) & (\varphi_n,\varphi_1) & \cdots & (\varphi_n,\varphi_n) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} (f,\varphi_0) \\ (f,\varphi_1) \\ \vdots \\ (f,\varphi_n) \end{bmatrix}$$

$G_{ij} = (x_i, x_j)$ $\det(G_{ij}) \neq 0 \Leftrightarrow x_1, \cdots, x_n$ 线性无关

$S^*(x) = \sum_{k=0}^{n} a_k^* \varphi_k(x) \Rightarrow (f(x) - S^*(x), S(x)) = 0$ $\|f-S^*\|_2 \le \|f-S\|_2$

正交多项式逼近 $(\varphi_i, \varphi_j) = \begin{cases} 0, i \neq j \\ A_j > 0, i = j \end{cases}$

$a_k^* = \frac{(f,\varphi_k)}{(\varphi_k,\varphi_k)}$, $k=0,1,\cdots,n$ $S_n^*(x) = \sum_{j=0}^{n} \frac{(f,\varphi_j)}{\|\varphi_j\|^2} \varphi_j$

$\varphi_0(x) = 1$ $\varphi_k(x) = \varphi_k(x) - \sum_{j=0}^{k-1} \frac{(\varphi_k,\varphi_j)}{(\varphi_j,\varphi_j)} \varphi_j(x)$

1 x $\frac{1}{2}(3x^2-1)$ $\frac{1}{2}(5x^3-3x)$ ⋯

Legendre 多项式 $[-1,1]$ $\rho=1$ $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}[(x^2-1)^n]$

$\int_{-1}^{1} P_n(x) P_m(x) = \begin{cases} 0, n \neq m \\ \frac{2}{2n+1}, n = m \end{cases}$ $P(1) = 1$

Chebyshev 多项式 $[-1,1]$ $\rho = \frac{1}{\sqrt{1-x^2}}$ $T_n(x) = \cos(n \arccos x)$

$(T_n, T_m) = \begin{cases} 0, m \neq n \\ \frac{\pi}{2}, m = n \neq 0 \\ \pi, m = n = 0 \end{cases}$ 零点 $x_k = \cos\frac{(2k-1)\pi}{2n}$, $k=1,2,\cdots,n$

Laguerre 多项式 $[0,+\infty)$ $\rho = e^{-x}$ $L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x})$

Hermite 多项式 $(-\infty,+\infty)$ $\rho = e^{-x^2}$ $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$

线性最小二乘 $\|\vec{b} - A\vec{x}^*\|_2 = \min_{\vec{x}} \|A\vec{x} - \vec{b}\|_2 \Leftrightarrow A^T A \vec{x}^* = A^T \vec{b}$

Pade 逼近 $R_{n,m} = \frac{a_0 + a_1 x + \cdots + a_n x^n}{1 + b_1 x + \cdots + b_m x^m} = \frac{P_n(x)}{Q_m(x)}$ $f(x) - R_{n,m} = O(x^N)$ $N = n+m+1$

$f(x) Q_m(x) - P_n(x) = \sum_{j=n+m+1}^{\infty} c_j x^j$

$\sqrt{x}$ $\frac{1}{\sqrt{x}}$ e^{-2x}

1 x x^2+ax+b

数值积分

中点公式 $\int_a^b f(x)dx \approx f(\frac{a+b}{2})(b-a)$

梯形公式 $\int_a^b f(x)dx \approx \frac{f(a)+f(b)}{2}(b-a)$

插值型求积 $\int_a^b f(x)dx \approx \sum_{k=0}^{n} A_k f(x_k)$

A_k：求积系数　x_k：求积节点

$E_n(x^k)=0,\ k=0,1,\cdots,m,\ E_n(x^{m+1})\neq 0 \Rightarrow$ m次代数精度

Gauss型求积 $I(f)=\int_a^b \rho(x)f(x)dx=\sum_{k=0}^{n} A_k f(x_k)$ 有 $2n+1$ 次代数精度

$\Leftrightarrow x_0,\cdots,x_n$ 为 $[a,b]$ 上权为 $\rho(x)$ 的 $n+1$ 次正交多项式零点

ODE数值解

$\frac{dy}{dx}=f(x,y)$，$y(a)=y_0$

向前欧拉 $y_{n+1}=y_n+hf(x_n,y_n)$，$y_0=y(a)$

向后欧拉 $y_{n+1}=y_n+hf(x_{n+1},y_{n+1})$

显式单步法 $y_{n+1}=y_n+h\varphi(x_n,y_n,h)$ → 增量函数

截断误差 $T_{n+1}(x)=y(x_{n+1})-y(x_n)-h\varphi(x_n,y(x_n),h)$

$T(x,y;h)=y(x+h)-y(x)-h\varphi(x,y(x),h)=A(x)h^p+O(h^{p+1}) \Rightarrow p$ 阶

局部截断误差 $T_{n+1}(x_n,y(x_n),h)=A(x_n)h^p+O(h^{p+1})$ 主局部截断误差 $A(x_n)h^p$

$\varphi(x,y,0)=f(x,y)$ 相容 $\Leftrightarrow$ 方法至少一阶

收敛性：$\forall x_n=x_0+nh$，$\lim_{h\to 0} y_n=y(x_n)$

（φ 对 x,y,h 连续，对 y 满足Lip条件）单步法相容 $\Leftrightarrow$ 单步法收敛 $y(x_n)-y_n=O(h^p)$

稳定性：$\forall$ 初始 e_0，$\exists\ 0<K<\infty$、$h_0>0$，当 $0<h\le h_0$ 时，$|e_n|\le K|e_0|$

绝对稳定性：试验方程 $y'=\lambda y=f$　$|\frac{y_{n+1}}{y_n}|<1$：绝对稳定区域 R_A

R_A 与实轴交集：绝对稳定区间

$y'(x)=f \quad y''(x)=\frac{\partial f}{\partial x}+f\frac{\partial f}{\partial y}$

$y'''(x)=\frac{\partial^2 f}{\partial x^2}+2f\frac{\partial^2 f}{\partial x\partial y}+f^2\frac{\partial^2 f}{\partial y^2}+\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}+f\left(\frac{\partial f}{\partial y}\right)^2$

S级 Runge-Kutta方法：$y_{n+1}=y_n+h\varphi(x_n,y_n;h)$

$\varphi(x_n,y_n,h)=\sum_{i=1}^{S}b_ik_i \quad k_i=f(x_n+c_ih,\ y_n+h\sum_{j=1}^{S}a_{ij}k_j)$

$=\sum_{j=1}^{S}a_{ij}$

线性多步法 $\sum_{i=0}^{k}\alpha_i y_{n+i}=h\sum_{i=0}^{k}\beta_i f_{n+i}$，$\alpha_k=1$，$\alpha_0^2+\beta_0^2\neq 0$

$\alpha_0 y_n+\alpha_1 y_{n+1}+\cdots+\alpha_k y_{n+k}=h(\beta_0 f_n+\beta_1 f_{n+1}\cdots+\beta_k f_{n+k})$

$\mathcal{L}(y(t);h)=\sum_{i=0}^{k}[\alpha_i y(t+ih)-h\beta_i y'(t+ih)]=O(h^{p+1})$：p阶

第一特征多项式 $\rho(\xi)=\sum_{j=0}^{k}\alpha_j\xi^j$ 第二特征多项式 $\sigma(\xi)=\sum_{j=0}^{k}\beta_j\xi^j$

多步法相容 $\Leftrightarrow T(t_i,y(t_i);h)=o(h)\Leftrightarrow \rho(1)=0,\ \rho'(1)=\sigma(1)$

根条件：$\rho(\xi)$的k个根模不大于1，模等于1的根为单根

多步法收敛 $\Rightarrow$ 根条件　根条件 $\Leftrightarrow$ 稳定性　多步法相容 $\Rightarrow$ (根条件 $\Leftrightarrow$ 收敛)